Learning Outcomes

In this chapter, you'll learn…

- how to calculate work done by a system when its volume changes.
- how to interpret and use the first law of thermodynamics.
- four important kinds of thermodynamic processes.
- why the internal energy of an ideal gas depends on temperature only.
- the difference between molar heat capacities at constant volume and at constant pressure.
- how to analyze adiabatic processes in an ideal gas.

Introduction

- A steam locomotive operates using the laws of thermodynamics, but so do air conditioners and car engines.

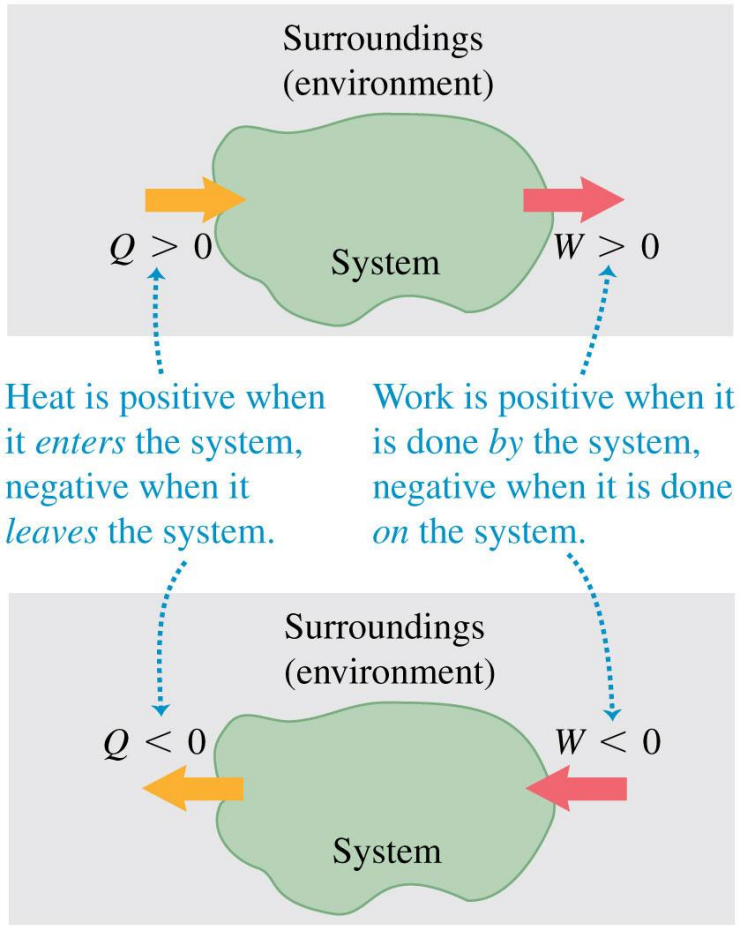
- We shall revisit the conservation of energy in the form of the first law of thermodynamics.

Thermodynamics Systems (1 of 2)

- A **thermodynamic system** is any collection of objects that may exchange energy with its surroundings.
- The popcorn in the pot is a thermodynamic system.
- In the thermodynamic process shown here, heat is added to the system, and the system does work on its surroundings to lift the lid of the pot.

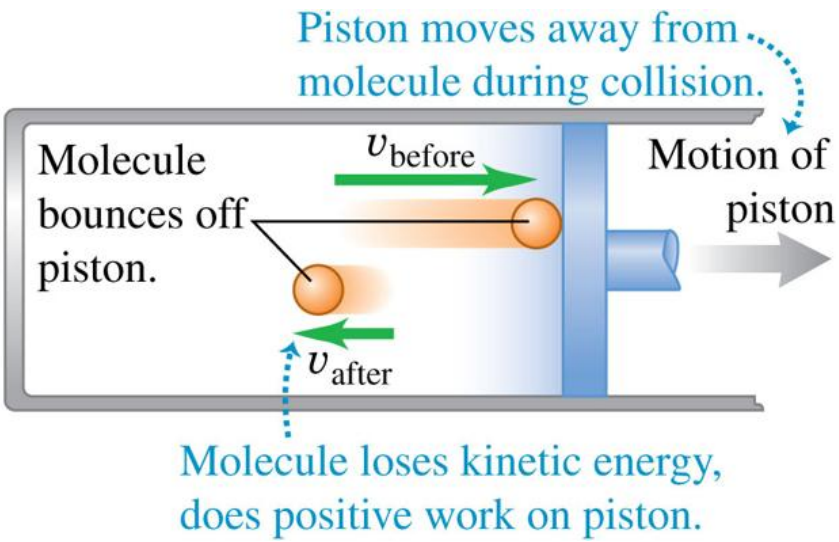
Thermodynamics Systems (2 of 2)

- In a **thermodynamic process**, changes occur in the state of the system.
- Careful of signs!
- *Q* is positive when heat flows **into** a system.
- *W* is the work done **by** the system, so it is positive for expansion.



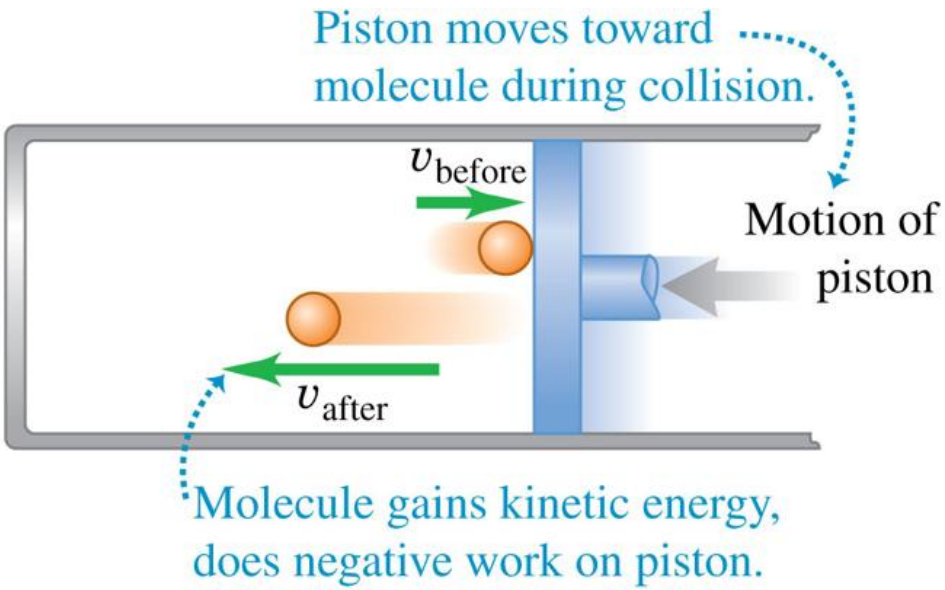
Work Done During Volume Changes (1 of 3)

- We can understand the work done by a gas in a volume change by considering a molecule in the gas.
- When one such molecule collides with a surface moving to the right, so the volume of the gas increases, the molecule does **positive** work on the piston.



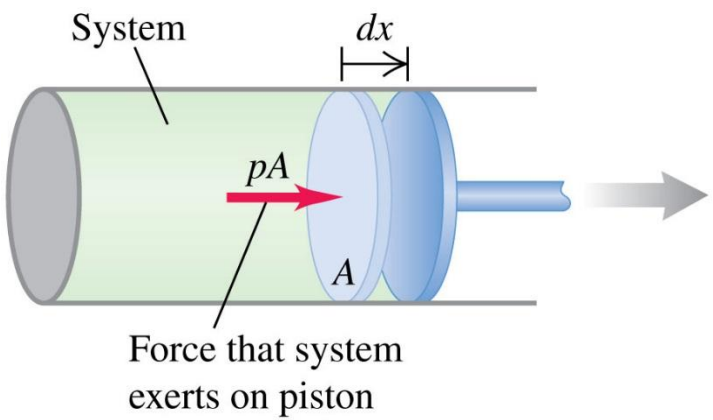
Work Done During Volume Changes (2 of 3)

- If the piston moves toward the left as in the figure shown here, so the volume of the gas decreases; positive work is done on the molecule during the collision.
- Hence the gas molecules do **negative** work on the piston.



Work Done During Volume Changes (3 of 3)

- The infinitesimal work done by the system during the small expansion dx is $dW = pA\,dx$.
- In a finite change of volume from V_1 to V_2:



Work done in a volume change $\rightarrow$ $$W = \int_{V_1}^{V_2} p\,dV$$

Upper limit = final volume

Integral of the pressure with respect to volume

Lower limit = initial volume

- Video Tutor Solution: Example 19.1

Question 1

10 pts

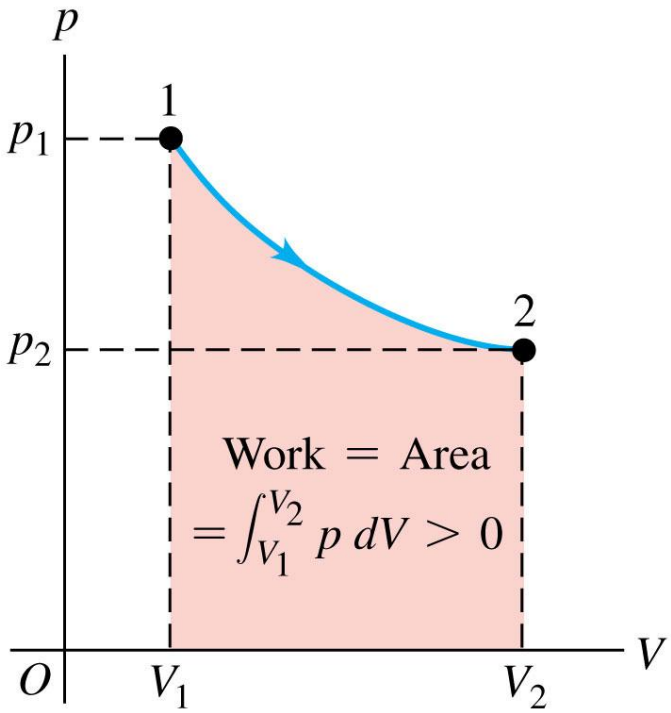
A system of gas decreases in volume from 2 m^3 to 0.8 m^3 at a constant pressure of 65 kPa. Calculate the work done **by** the gas during this process.

W = _____kJ

Correct Answer -78

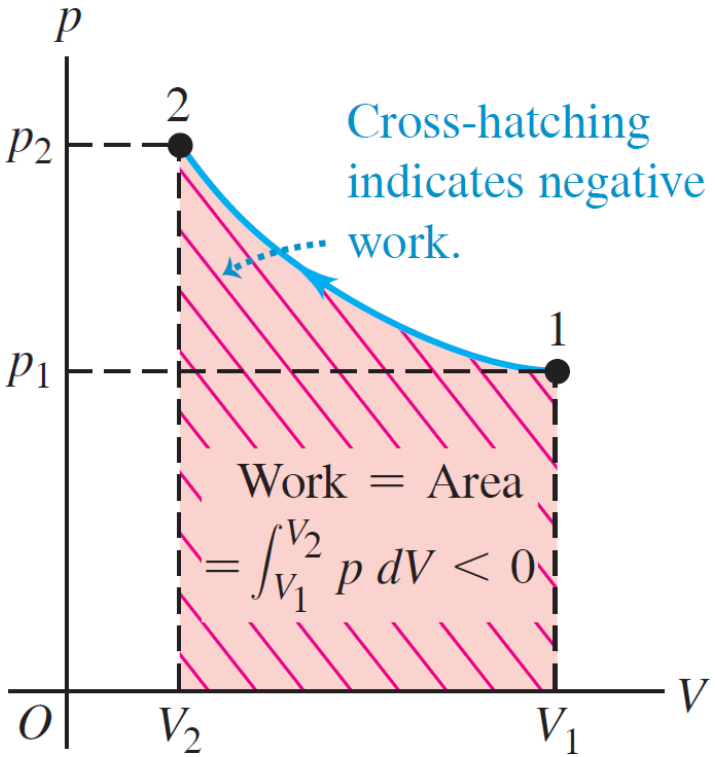
Work on a *pV*-Diagram (1 of 3)

- The work done equals the area under the curve on a *pV*-diagram.
- Shown in the graph is a system undergoing an **expansion** with varying pressure.



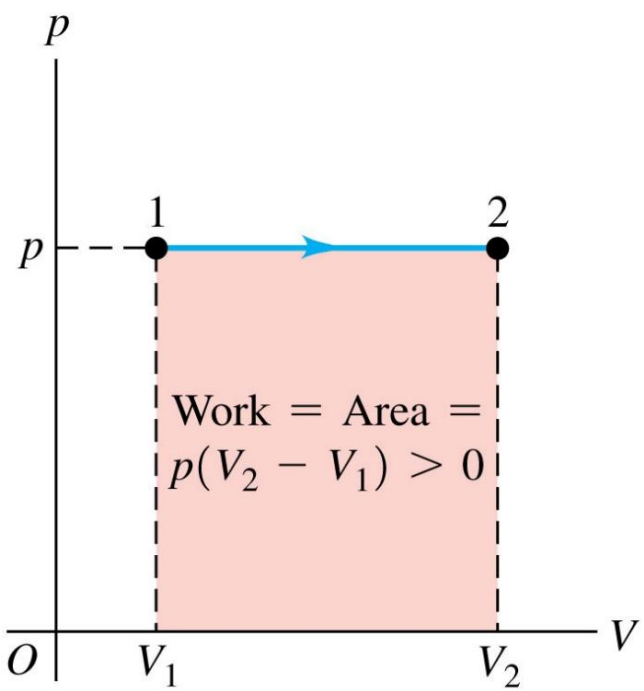
Work on a *pV*-Diagram (2 of 3)

- Shown in the graph is a system undergoing a **compression** with varying pressure.
- In this case the work is **negative**.



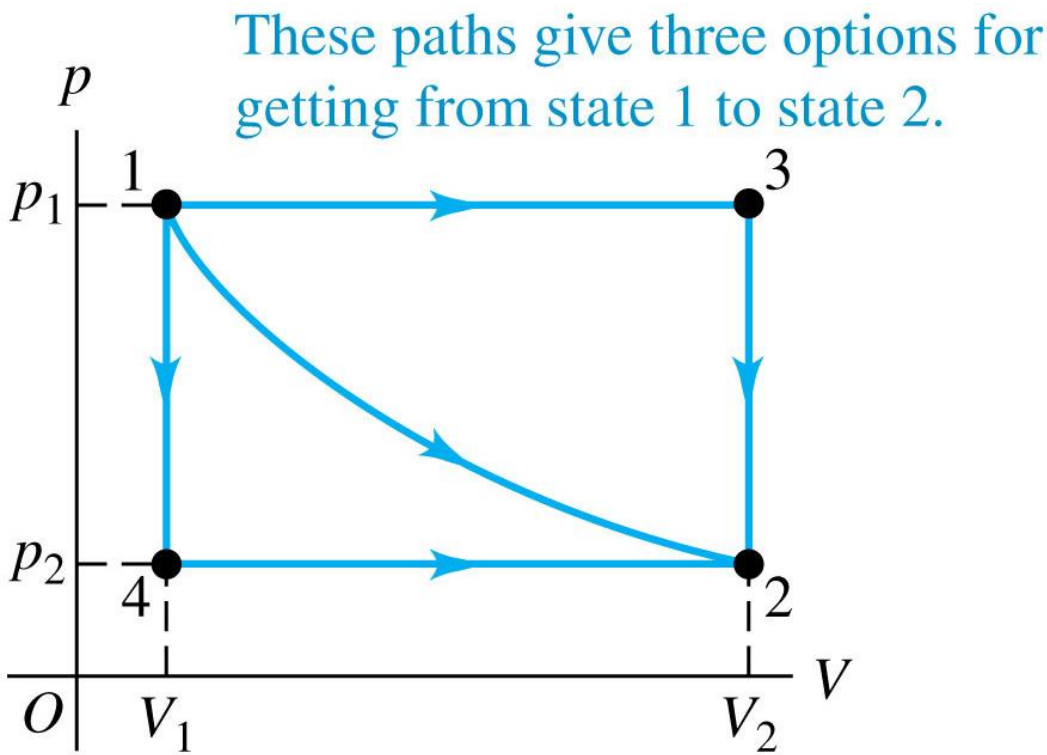
Work on a *pV*-Diagram (3 of 3)

- Shown in the graph is a system undergoing an expansion with **constant** pressure.
- In this case, $W = p(V_2 - V_1)$



Work Depends on the Path Chosen (1 of 4)

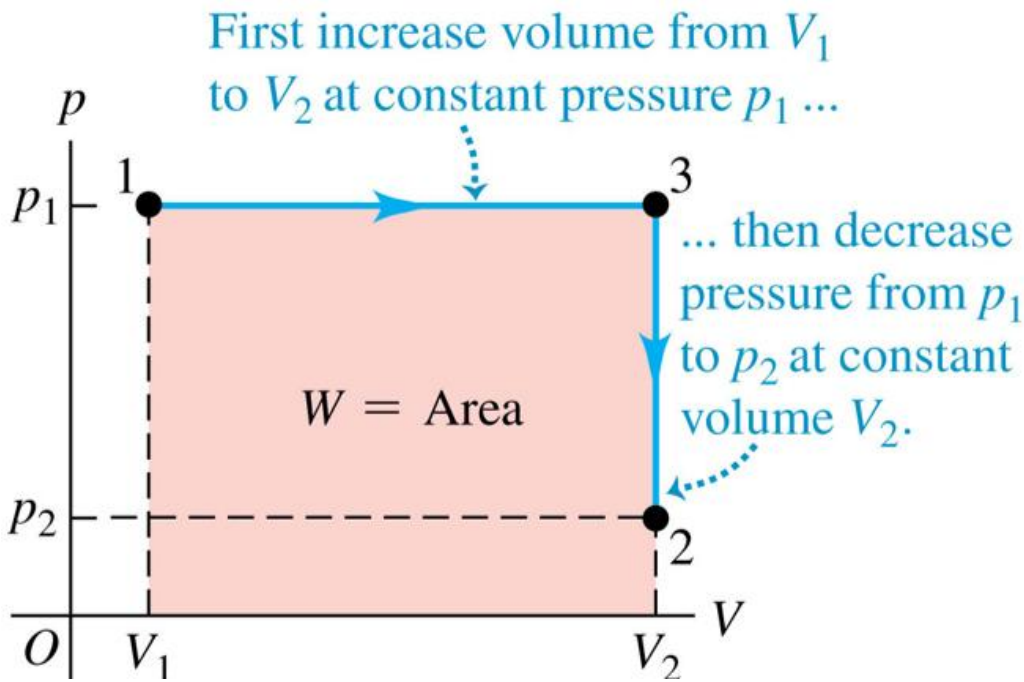
- Consider three different paths on a *pV*-diagram for getting from state 1 to state 2.



Work Depends on the Path Chosen (2 of 4)

- The system does a large amount of work under the path

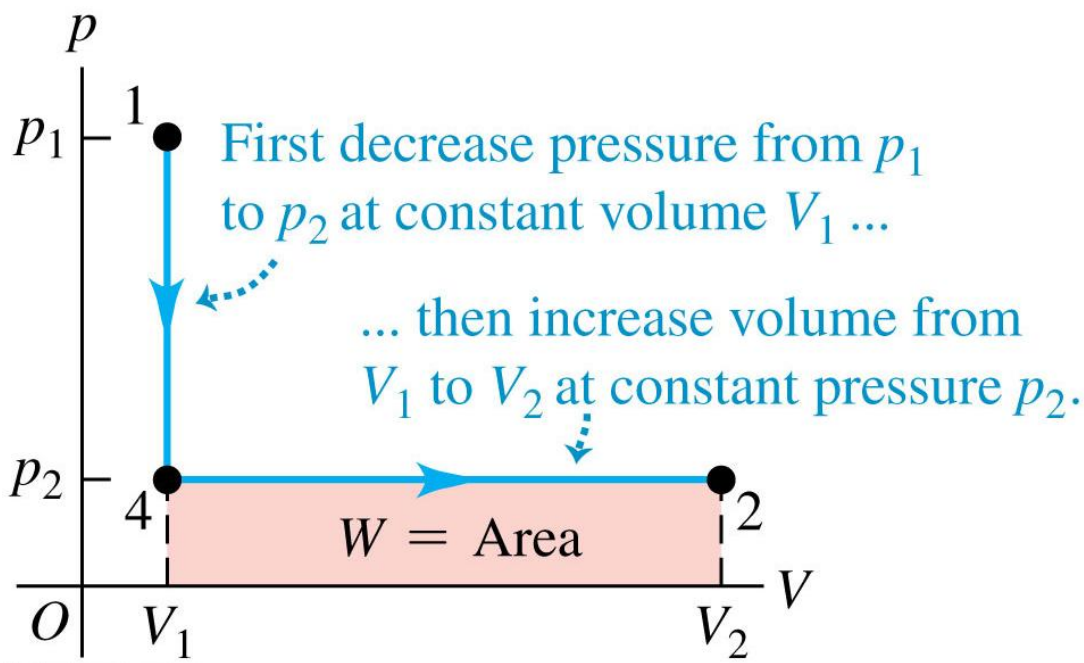
$$1 \rightarrow 3 \rightarrow 2.$$



Work Depends on the Path Chosen (3 of 4)

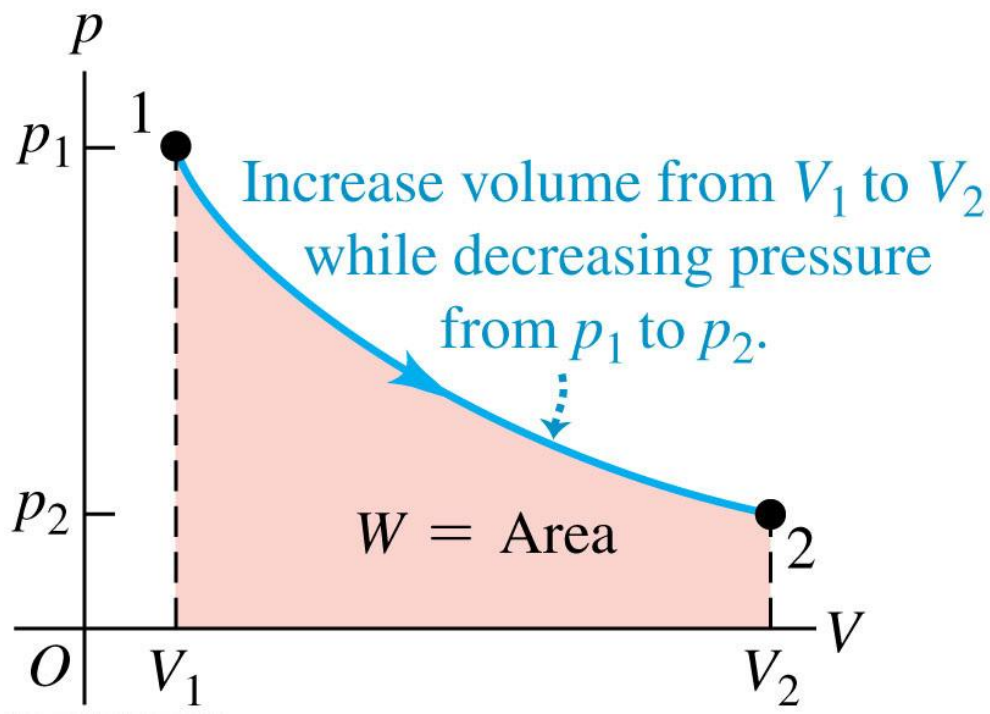
- The system does a small amount of work under the path

$$1 \rightarrow 4 \rightarrow 2.$$



Work Depends on the Path Chosen (4 of 4)

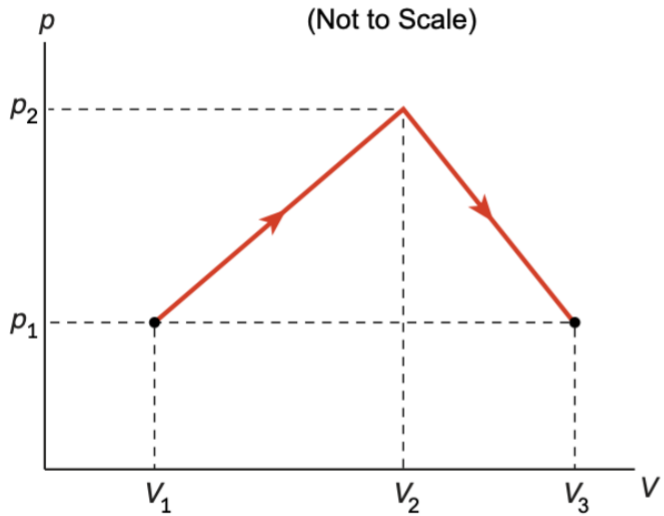
- Along the smooth curve from 1 to 2, the work done is different from that for either of the other paths.



Question 2

10 pts

In the given pV-diagram (not to scale), p_1 = 109 kPa, p_2 = 459 kPa; V_1 = 1.9 m^3, V_2 = 4.5 m^3, and V_3 = 5.3 m^3. Calculate the work in kJ done **on** the system during these processes.



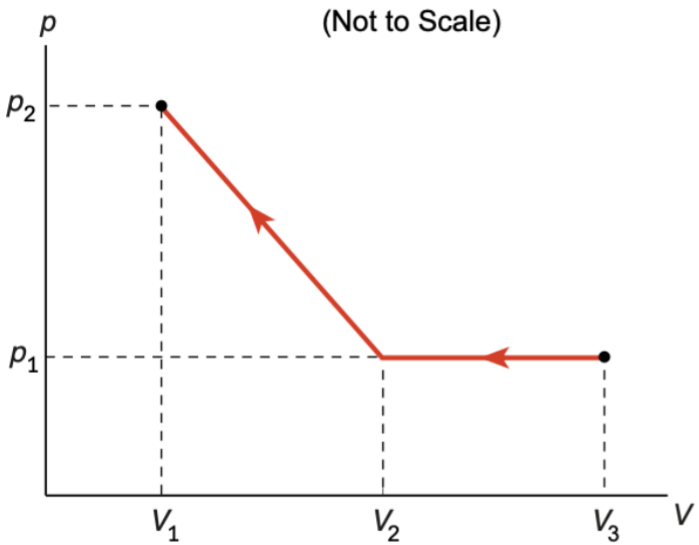
W_{on} = _____kJ

Correct Answer -965.6

Question 3

10 pts

In the given pV-diagram (not to scale), p_1 = 84.6 kPa, p_2 = 222.1 kPa; V_1 = 1 m^3, V_2 = 1.9 m^3, and V_3 = 3.4 m^3. Calculate the work in kJ done **on** the system during these processes.



W_{on} = _____kJ

Correct Answer 264.915

First Law of Thermodynamics (1 of 4)

- The change in the internal energy *U* of a system is equal to the heat added minus the work done by the system:

First law of thermodynamics:

Internal energy change of thermodynamic system

$$\Delta U = Q - W$$

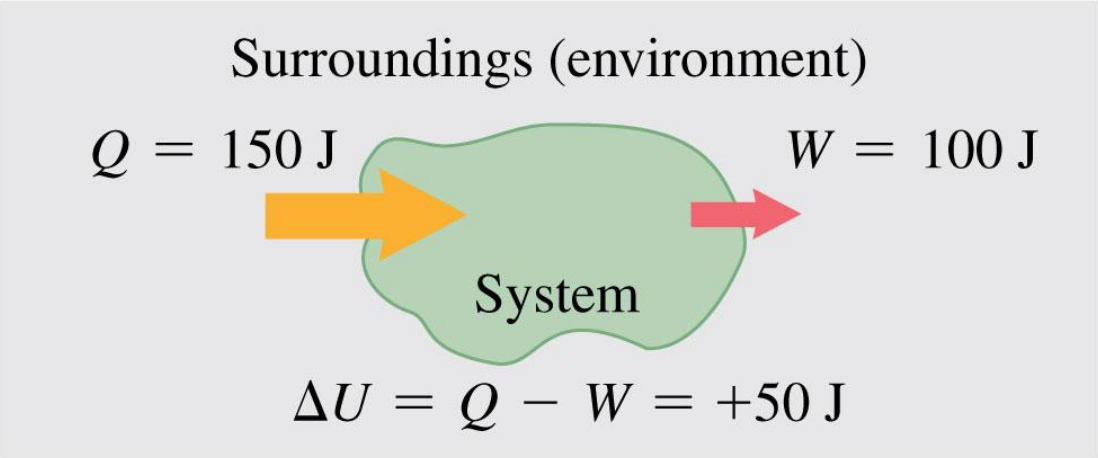
Heat added to system

Work done by system

- The first law of thermodynamics is just a generalization of the conservation of energy.
- Both *Q* and *W* depend on the path chosen between states, but is **independent of the path**.
- If the changes are infinitesimal, we write the first law as $dU = dQ - dW$.
- Video Tutor Solution: Example 19.3

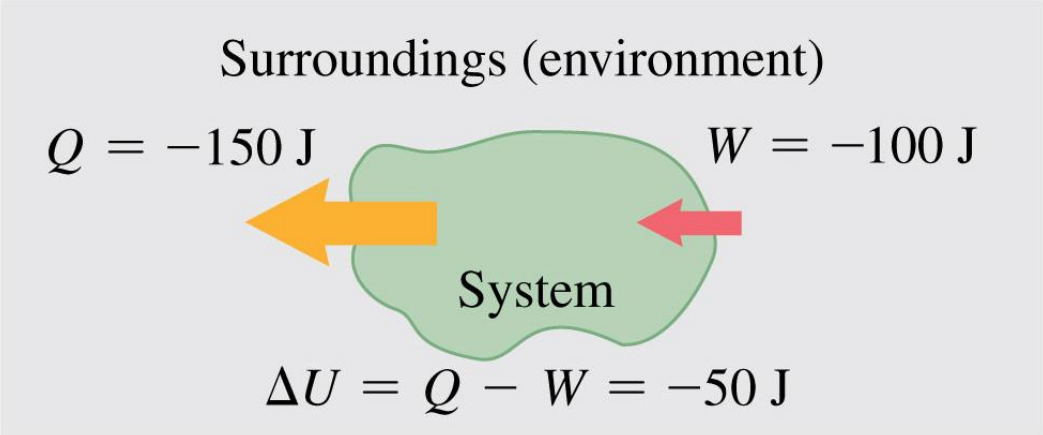
First Law of Thermodynamics (2 of 4)

- In a thermodynamic process, the internal energy *U* of a system may **increase**.
- In the system shown below, more heat is added to the system than the system does work.
- So the internal energy of the system increases.



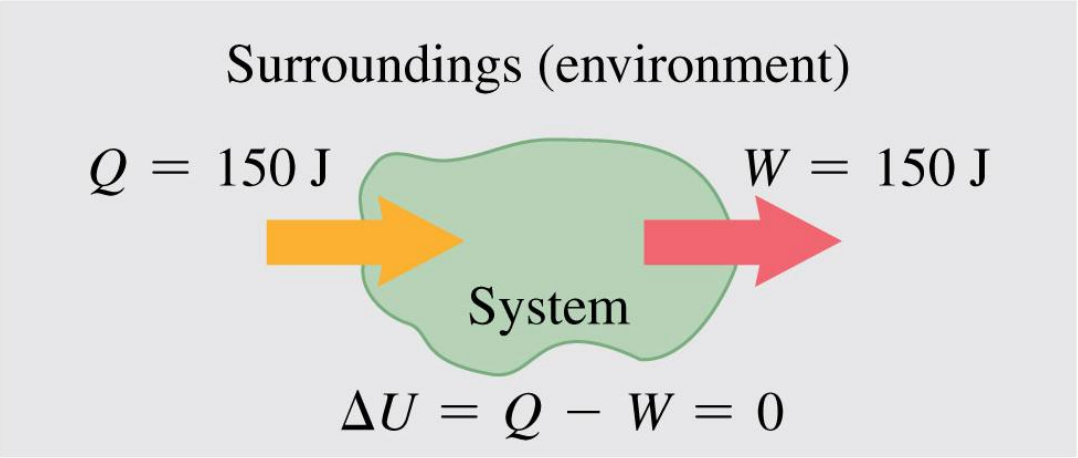
First Law of Thermodynamics (3 of 4)

- In a thermodynamic process, the internal energy *U* of a system may **decrease**.
- In the system shown below, more heat flows out of the system than work is done.
- So the internal energy of the system decreases.



First Law of Thermodynamics (4 of 4)

- In a thermodynamic process, the internal energy *U* of a system may **remain the same**.
- In the system shown below, the heat added to the system equals the work done by the system.
- So the internal energy of the system is unchanged.



First law of exercise Thermodynamics

- Your body is a thermodynamic system.
- When you do a push-up, your body does work, so $W > 0$.
- Your body also warms up during exercise; by perspiration and other means the body rids itself of this heat, so $Q < 0$.

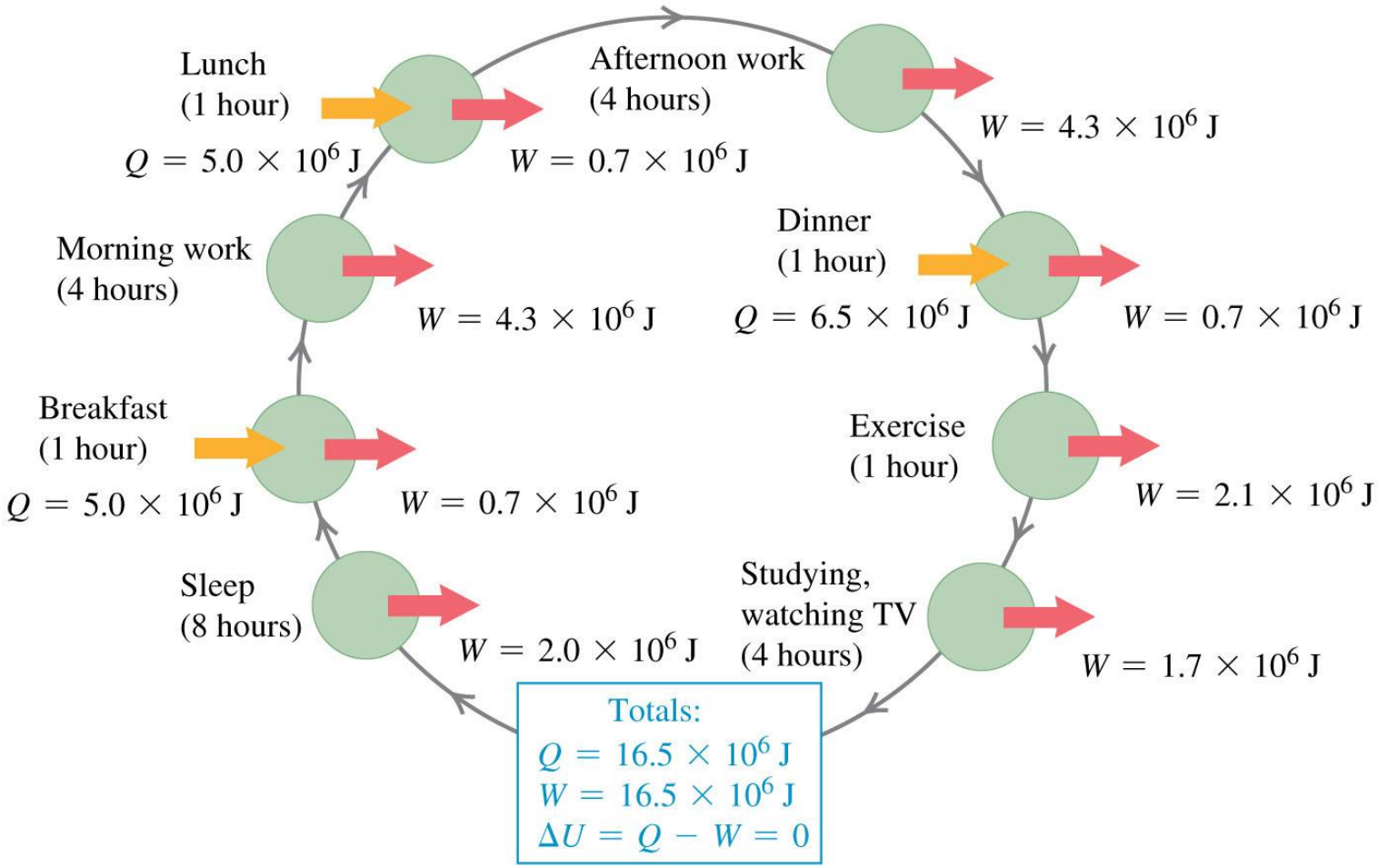
- Since Q is negative and W is positive, $U = Q - W < 0$ and the body's internal energy decreases.
- That's why exercise helps you lose weight: It uses up some of the internal energy stored in your body in the form of fat.

Internal Energy

- The internal energy of a cup of coffee depends on just its thermodynamic state—how much water and ground coffee it contains, and what its temperature is.

- It does not depend on the history of how the coffee was prepared—that is, the thermodynamic path that led to its current state.

Cyclic Thermodynamic Process for a Human Day

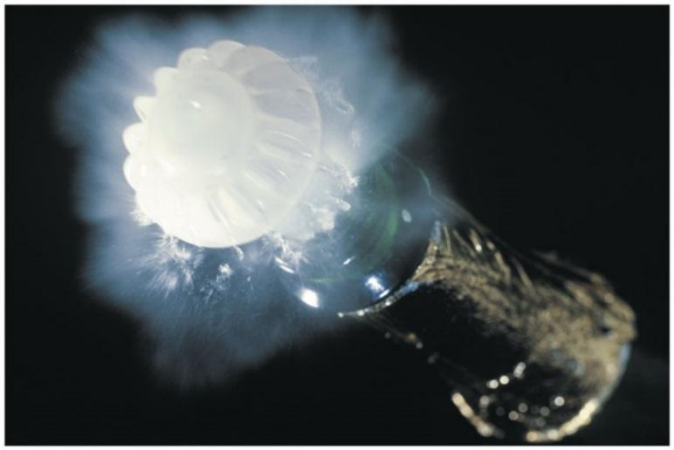


Four Kinds of Thermodynamic Processes

- There are four specific kinds of thermodynamic processes that occur often in practical situations:
  - **Adiabatic**: No heat is transferred into or out of the system, so $Q = 0$. Also, $U_2 - U_1 = W$.
  - **Isochoric**: The volume remains constant, so $W = 0$.
  - **Isobaric**: The pressure remains constant, so $W = p(V_2 - V_1)$.
  - **Isothermal**: The temperature remains constant.

Adiabatic Process

- When the cork is popped on a bottle of champagne, the pressurized gases inside the bottle expand rapidly and do positive work on the outside air.
- There is little time for the gases to exchange heat with their surroundings, so the expansion is nearly adiabatic.
- Hence the internal energy of the expanding gases decreases $(\Delta U = -W)$ and their temperature drops.
- This makes water vapor condense and form a miniature cloud.

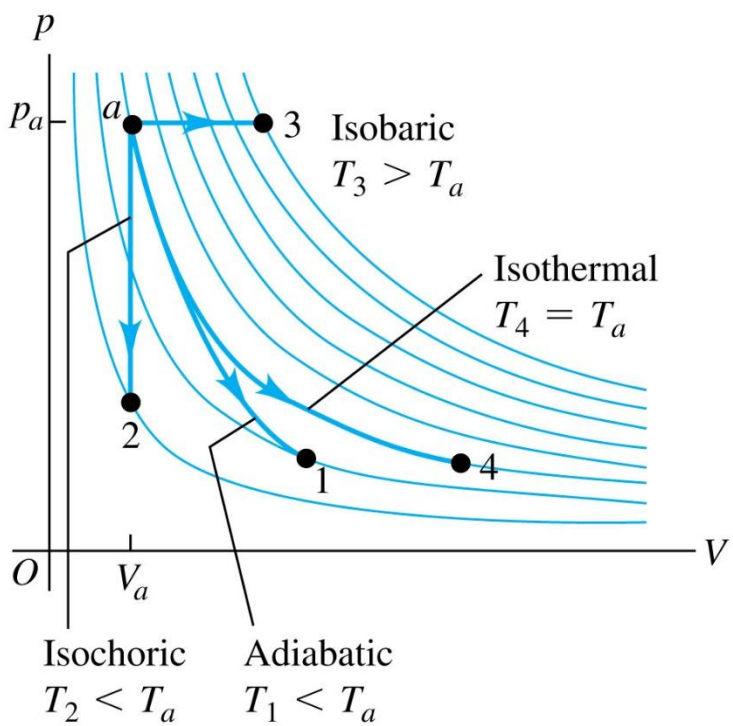


Isobaric Process

- Most cooking involves isobaric processes.
- That's because the air pressure above a saucepan or frying pan, or inside a microwave oven, remains essentially constant while the food is being heated.

The Four Processes on a pV-Diagram

- Shown are the paths on a *pV*-diagram for all four different processes for a constant amount of an ideal gas, all starting at state *a*.



Question 4

10 pts

Suppose a system of gas absorbs 741 J of energy while doing 0.24 times that amount of work on the environment. Calculate the change in the internal energy of the gas.

Correct Answer 563.16

Question 5

10 pts

An ideal monatomic gas increases in pressure from 119 kPa to 210 kPa at a constant volume of 1.4 m^3. Calculate the energy in kJ transferred by heat during the process.

Correct Answer 191.1

Question 6

10 pts

An ideal monatomic gas increases in pressure from 150 kPa to 206 kPa at a constant volume of 1.8 m^3. Calculate the change in the internal energy in kJ.

Correct Answer 151.2

Question 7

10 pts

A monatomic ideal gas expands at constant pressure of 123 kPa from 0.7 m^3 to 2.9 m^3. Calculate the work done **by** the gas in kJ.

Correct Answer 270.6

Question 8

10 pts

A system of diatomic ideal gas contracts from a volume of 3.5 m^3 to a final volume of 0.8 m^3. If the process proceeds at constant pressure of 168 kPa, calculate the energy in kJ transferred by heat, where positive is energy transferred into the system, and negative is energy transferred out of the system.

Correct Answer -1,587.6

Question 9

10 pts

A diatomic ideal gas undergoes an isothermal expansion from a volume of 1.1 m^3 and pressure 85 kPa to a final volume of 3 m^3. Calculate the change in the internal energy in kJ.

Correct Answer 0

Question 10

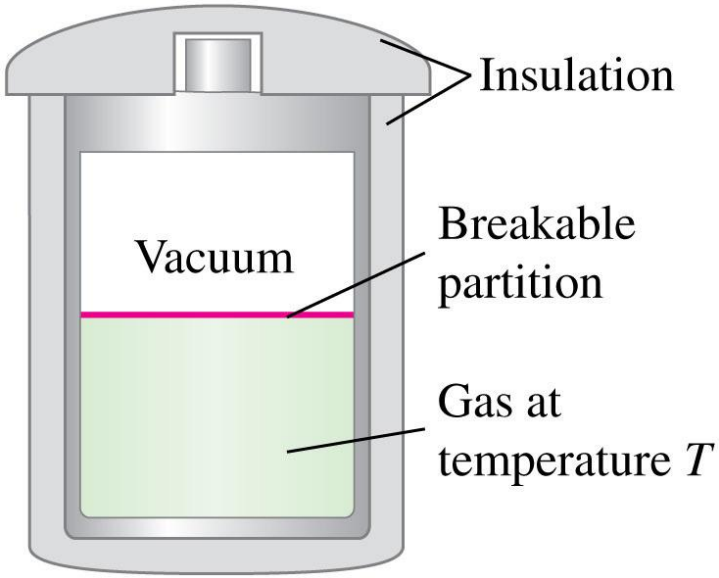
10 pts

A monatomic ideal gas undergoes an isothermal compression from a volume of 3.2 m^3 and pressure 71 kPa to a final volume of 1.3 m^3. Calculate the work in kJ done **by** the gas on its environment.

Correct Answer -204.6587

Internal Energy of an Ideal Gas

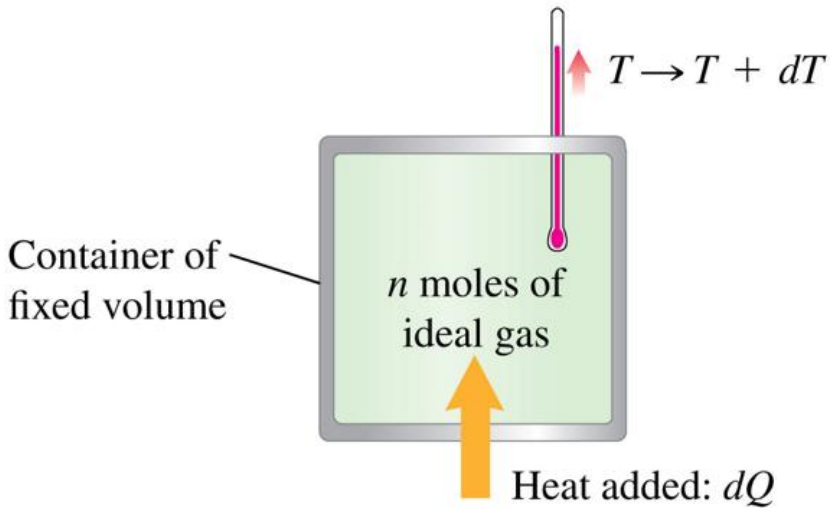
- The internal energy of an ideal gas depends **only** on its **temperature**, not on its pressure or volume.
- The temperature of an ideal gas does **not** change during a free expansion.



Heat Capacities of an Ideal Gas (1 of 2)

- C_V is the molar heat capacity at constant volume.
- To measure C_V, we raise the temperature of an ideal gas in a rigid container with constant volume, ignoring its thermal expansion.

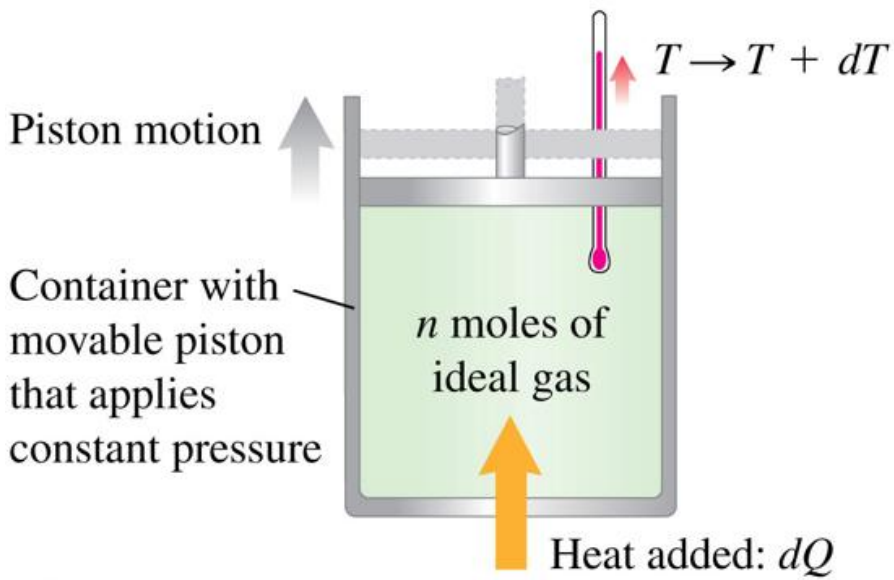
Constant volume: $dQ = nC_V\,dT$



Heat Capacities of an Ideal Gas (2 of 2)

- C_p is the molar heat capacity at constant pressure.
- To measure C_p, we let the gas expand just enough to keep the pressure constant as the temperature rises.

Constant pressure: $dQ = nC_p\,dT$

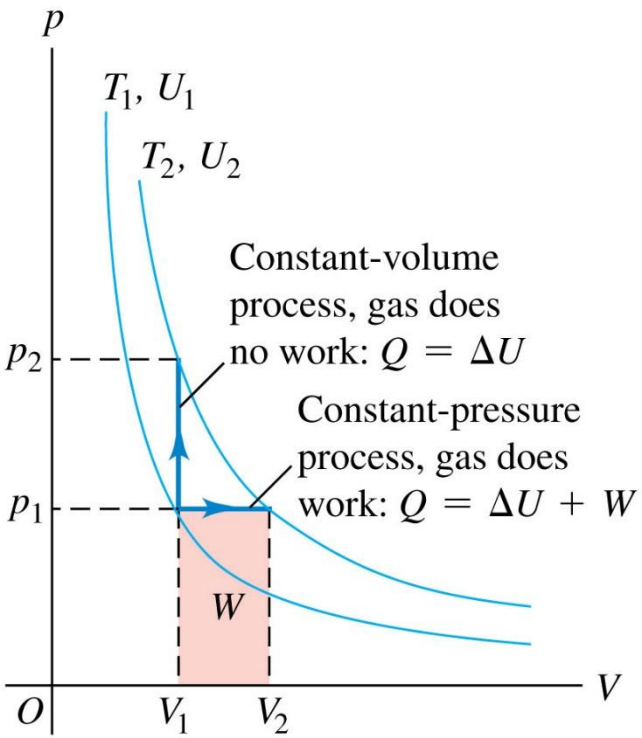


Relating *Cp* and *CV* for an Ideal Gas

- To produce the same temperature change, more heat is required at constant pressure than at constant volume since

  ΔU is the same in both cases.
- This means that $C_p > C_V$.
- $C_p = C_V + R$.
- R is the gas constant

$$R = 8.314 \frac{\text{J}}{\text{mol}} \cdot K.$$



The Ratio of Heat Capacities

- The **ratio of heat capacities** is:

Ratio of heat capacities $\gamma = \dfrac{C_p}{C_V}$

C_p: Molar heat capacity at constant *pressure*

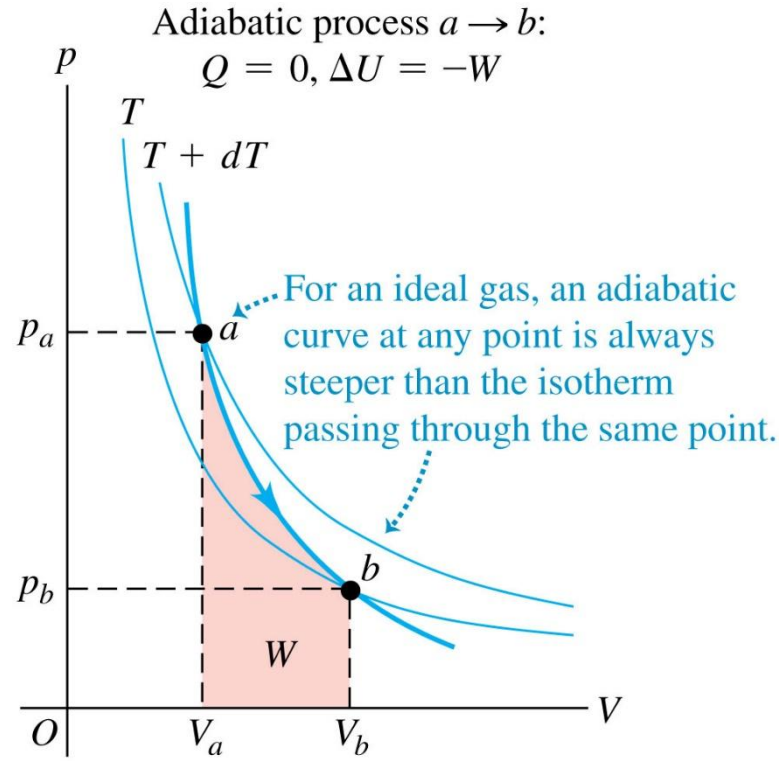
C_V: Molar heat capacity at constant *volume*

- For monatomic ideal gases, $\gamma = 1.67$.
- For diatomic ideal gases, $\gamma = 1.40$.

- Video Tutor Solution: Example 19.6

Adiabatic Processes for an Ideal Gas

- In an adiabatic process, no heat is transferred in or out of the gas, so $Q = 0$.
- Shown is a pV–diagram for an adiabatic expansion.
- As the gas expands, it does positive work W on its environment, so its internal energy decreases, and its temperature drops.



- Note that an adiabatic curve at any point is always steeper than an isotherm at that point.

Exhaling Adiabatically

- Put your hand a few centimeters in front of your mouth, open your mouth wide, and exhale.
- Your breath will feel warm on your hand, because the exhaled gases emerge at roughly the temperature of your body's interior.
- Now purse your lips as though you were going to whistle, and again blow on your hand.
- The exhaled gases will feel much cooler.
- In this case the gases undergo a rapid, essentially adiabatic expansion as they emerge from between your lips, so the temperature of the exhaled gases decreases.

Question 11

10 pts

A diatomic ideal gas is compressed adiabatically from 2.6 m^3 to 1 m^3. If the initial pressure is 65 kPa, calculate the final pressure in kPa.

Correct Answer 247.6715

Question 12

10 pts

A monatomic ideal gas increases in pressure adiabatically from an initial pressure of 1.00×10^5 Pa and volume of 3.00 m^3 to a final volume of 1.00 m^3. Two answers are required.

(a) Calculate the final pressure. Answer: Correct Answer 624 kPa

(b) Calculate the change in the internal energy. Answer: Correct Answer 486 kJ

Question 13

10 pts

A monatomic ideal gas expands adiabatically from 1.3 m^3 to 3 m^3. If the initial pressure is 111 kPa, calculate the energy in kJ transferred to or from the system by heat.

Correct Answer 0

Question 14

10 pts

On a *pV*-diagram, a diatomic gas process proceeds along the straight line from (2 m^3, 284 kPa) to (5 m^3, 415 kPa). Calculate the work done by the gas in kJ.

Correct Answer 1,048.5

Question 15

10 pts

State whether the following are always true, always false, or true sometimes.

(a) Energy is transferred by heat to a gas during an isobaric expansion.

Correct Answer Always True

(b) Energy is removed by heat from a gas during an isothermal process.

Correct Answer True sometimes

(c) Energy is added by heat during an adiabatic compression.

Correct Answer Always False

(d) Energy is transferred out of a gas during a constant volume pressure decrease.

Correct Answer Always True

Question 16

10 pts

Name the following thermodynamic processes by using one of the words below in each blank:

(a) The change in internal energy is zero: Correct Answer isothermal

(b) The work done by the system is zero: Correct Answer isochoric

(c) The energy transfer by heat is zero: Correct Answer adiabatic

Word list: adiabatic; isobaric; isothermal; isochoric; generic